

4 - OSPF

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Task 1 - Prepare your network topology

- Created Topology. Setup 2 adapters on all routes as Gigabyte Ethernet
 - Got red curved link - fixed it by changing the order of connecting nodes

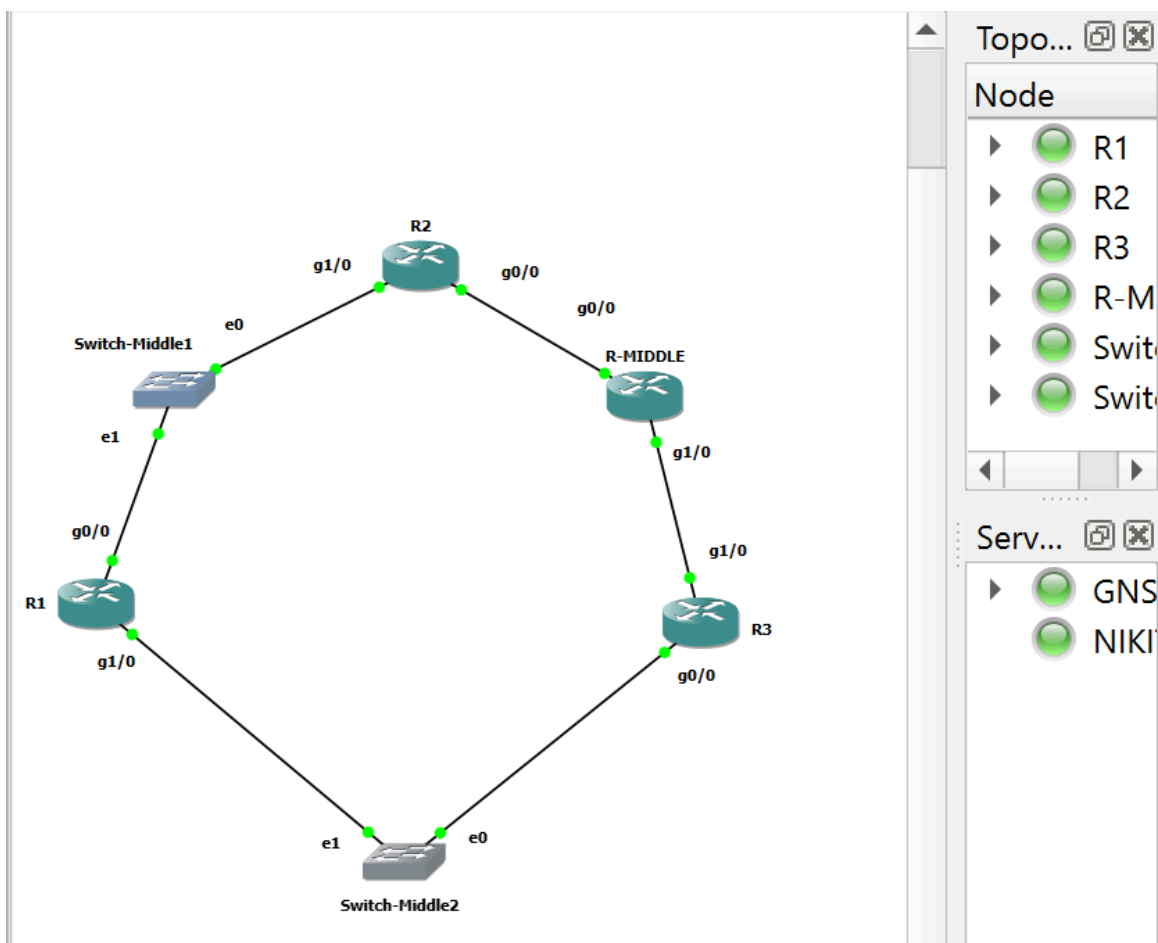


Figure. Topology II

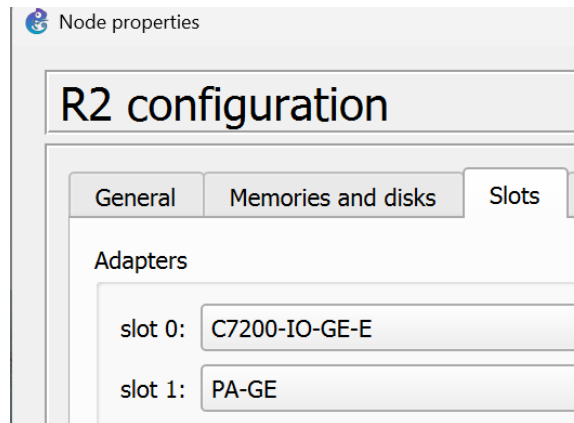


Figure. Configuration of adapters

- Setup settings on all routers. I prepared a simple network consisting of at least 3 routers, each one of them has a different subnet, and they should be able to reach each other. I did not write a static routes between different networks, but assigned static addresses!):

```
R2# conf t
R2 (config)# interface g0/0
R2 (config-if)# ip address 10.0.0.1 255.255.255.252
R2 (config-if)# no shut
R2 (config-if)# interface g1/0
R2 (config-if)# ip address 10.0.3.2 255.255.255.252
R2 (config-if)# no shut
```

```
R-MIDDLE# conf t
(config)# interface g0/0
(config-if)# ip address 10.0.0.2 255.255.255.252
(config-if)# no shut
(config-if)# interface g1/0
(config-if)# ip address 10.0.1.1 255.255.255.252
(config-if)# no shut
```

```

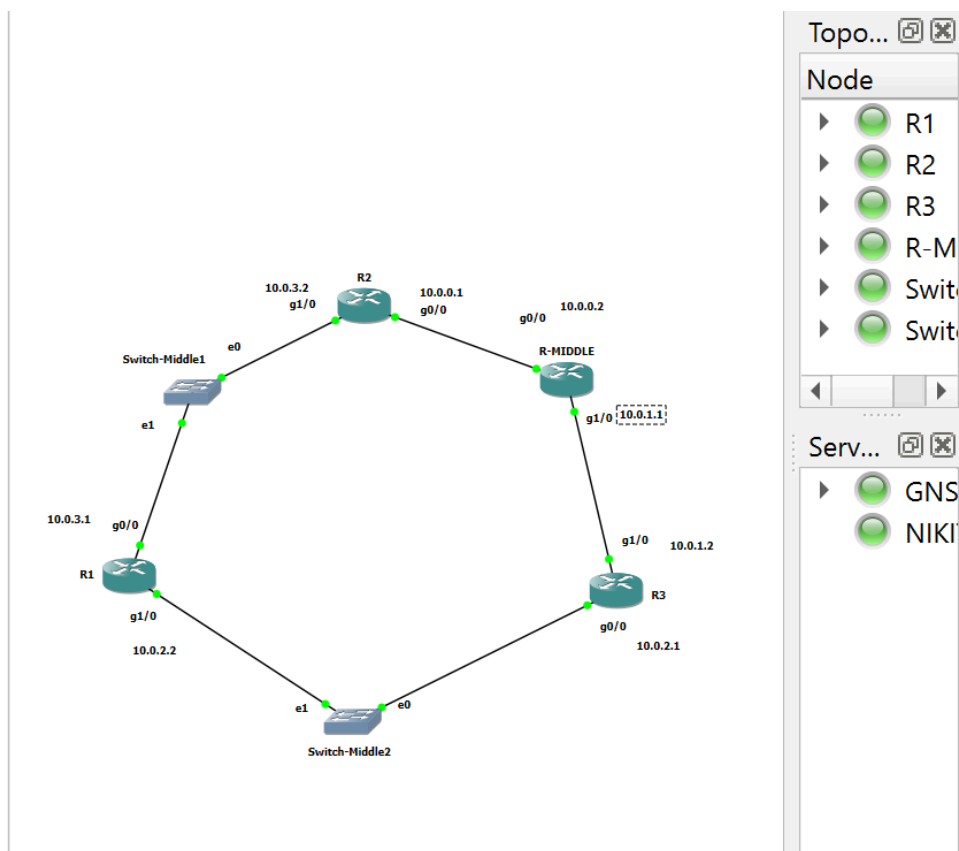
R3# conf t
(config-if)# interface g0/0
(config-if)# ip address 10.0.2.1 255.255.255.252
(config-if)# no shut
(config)# interface g1/0
(config-if)# ip address 10.0.1.2 255.255.255.252
(config-if)# no shut

```

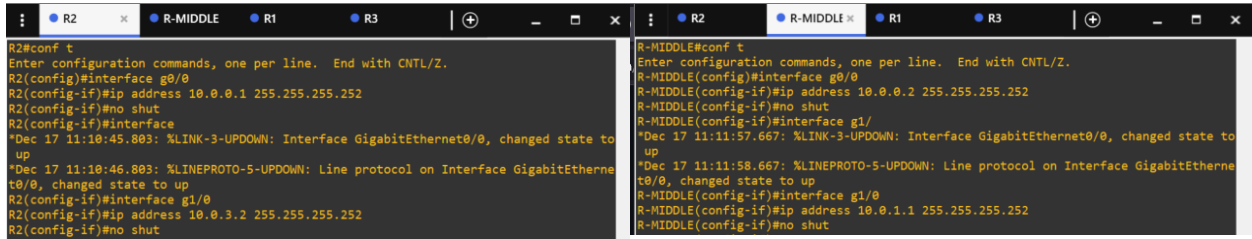
```

R1# conf t
(config)# interface g0/0
(config-if)# ip address 10.0.3.1 255.255.255.252
(config-if)# no shut
(config-if)# interface g1/0
(config-if)# ip address 10.0.2.2 255.255.255.252
(config-if)# no shut

```



Figures. Topology II with static IPS

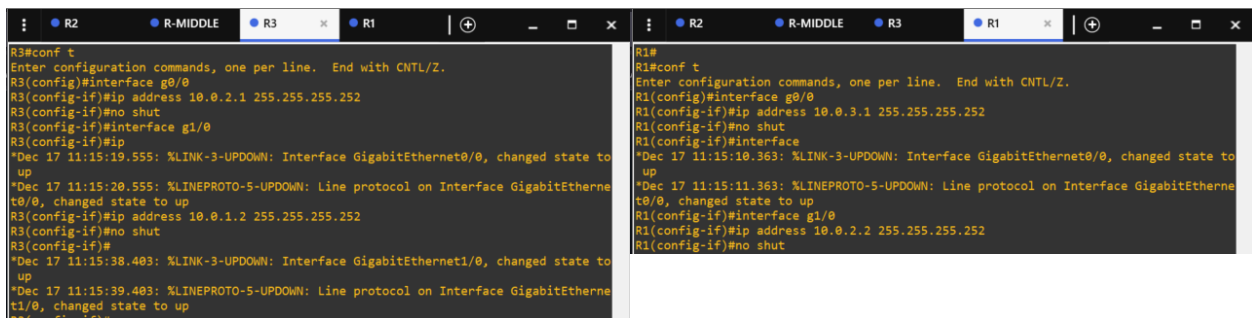


The image shows two terminal windows side-by-side. The left window is for R2 and the right is for R-MIDDLE. Both show the configuration of interfaces g0/0 and g1/0 with static IP addresses and the 'no shut' command.

```
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#interface g0/0
R2(config-if)#ip address 10.0.0.1 255.255.255.252
R2(config-if)#no shut
R2(config-if)#interface
*Dec 17 11:10:45.803: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to
up
*Dec 17 11:10:46.803: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthe
t0/0, changed state to up
R2(config-if)#interface g1/0
R2(config-if)#ip address 10.0.3.2 255.255.255.252
R2(config-if)#no shut

R-MIDDLE#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R-MIDDLE(config)#interface g0/0
R-MIDDLE(config-if)#ip address 10.0.0.2 255.255.255.252
R-MIDDLE(config-if)#no shut
R-MIDDLE(config-if)#interface g1/
*Dec 17 11:11:57.667: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to
up
*Dec 17 11:11:58.667: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthe
t0/0, changed state to up
R-MIDDLE(config-if)#interface g1/0
R-MIDDLE(config-if)#ip address 10.0.1.1 255.255.255.252
R-MIDDLE(config-if)#no shut
```

Figures. R2 and R-MIDDLE



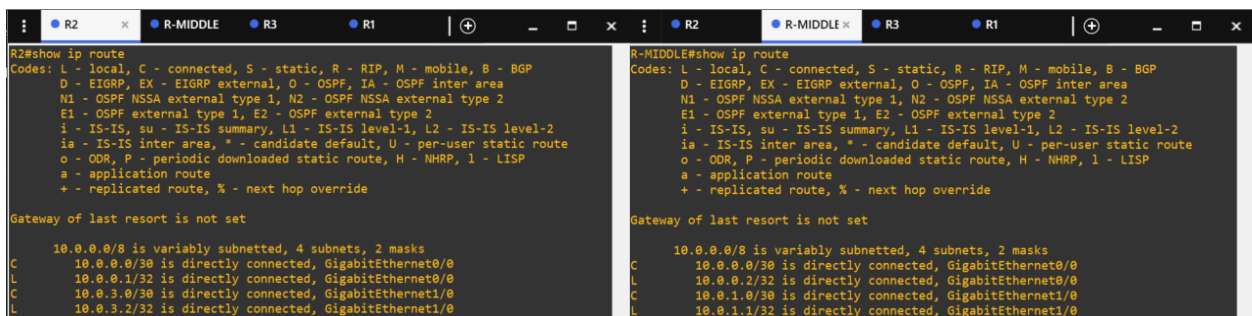
The image shows two terminal windows side-by-side. The left window is for R3 and the right is for R1. Both show the configuration of interfaces g0/0 and g1/0 with static IP addresses and the 'no shut' command.

```
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#interface g0/0
R3(config-if)#ip address 10.0.2.1 255.255.255.252
R3(config-if)#no shut
R3(config-if)#interface g1/0
R3(config-if)#ip
*Dec 17 11:15:19.555: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to
up
*Dec 17 11:15:20.555: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthe
t0/0, changed state to up
R3(config-if)#ip address 10.0.1.2 255.255.255.252
R3(config-if)#no shut
R3(config-if)#
*Dec 17 11:15:38.403: %LINK-3-UPDOWN: Interface GigabitEthernet1/0, changed state to
up
*Dec 17 11:15:39.403: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthe
t1/0, changed state to up

R1#
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface g0/0
R1(config-if)#ip address 10.0.3.1 255.255.255.252
R1(config-if)#no shut
R1(config-if)#interface
*Dec 17 11:15:10.363: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to
up
*Dec 17 11:15:11.363: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthe
t0/0, changed state to up
R1(config-if)#interface g1/0
R1(config-if)#ip address 10.0.2.2 255.255.255.252
R1(config-if)#no shut
```

Figures. R3 and R1

- Now I checked routes on nodes using `show ip route:`



The image shows two terminal windows side-by-side. The left window is for R2 and the right is for R-MIDDLE. Both show the output of the 'show ip route' command, displaying the routing table for each router.

```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP
a - application route
+ - replicated route, % - next hop override

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C    10.0.0.0/30 is directly connected, GigabitEthernet0/0
L    10.0.0.1/32 is directly connected, GigabitEthernet0/0
C    10.0.3.0/30 is directly connected, GigabitEthernet1/0
L    10.0.3.2/32 is directly connected, GigabitEthernet1/0

R-MIDDLE#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP
a - application route
+ - replicated route, % - next hop override

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C    10.0.0.0/30 is directly connected, GigabitEthernet0/0
L    10.0.0.2/32 is directly connected, GigabitEthernet0/0
C    10.0.1.0/30 is directly connected, GigabitEthernet1/0
L    10.0.1.1/32 is directly connected, GigabitEthernet1/0
```

Figures. R2 and R-MIDDLE

The image shows two terminal windows side-by-side. The left window is for R3 and the right window is for R1. Both show the output of the 'show ip route' command. The output includes a legend for route codes (L, C, S, R, M, B, D, N1, E1, i, ia, o, a, +) and a list of connected interfaces with their IP addresses and masks. For R3, the interfaces are 10.0.0.0/8, 10.0.1.0/30, 10.0.1.2/32, 10.0.2.0/30, and 10.0.2.1/32. For R1, the interfaces are 10.0.0.0/8, 10.0.2.0/30, 10.0.2.2/32, 10.0.3.0/30, and 10.0.3.1/32.

Figures. R3 and R1

- Nodes can not ping remote nodes, that are on hop+1, because OSPF is not configured yet. For example R-MIDDLE can ping R3 (10.0.1.2) and R2 (10.0.0.1), but R1 (10.0.3.1) is unreachable.

The image shows a terminal window for R-MIDDLE. It displays the results of three ping commands: 'ping 10.0.0.1', 'ping 10.0.1.2', and 'ping 10.0.3.1'. The first two pings are successful with an 80 percent success rate. The third ping to 10.0.3.1 fails with a 0 percent success rate.

Figure. Pinging from R-MIDDLE

Task 2 - OSPF Learning & Configuring

1. Deploy OSPF in your chosen network topology.

```
R2# conf t
(config)# router ospf 1
(config-router)# router-id 1.1.1.1
(config-router)# network 10.0.0.0 0.0.0.3 area 0
(config-router)# network 10.0.3.0 0.0.0.3 area 0
```

```
R-MIDDLE# conf t
(config)# router ospf 1
(config-router)# router-id 2.2.2.2
(config-router)# network 10.0.0.0 0.0.0.3 area 0
(config-router)# network 10.0.1.0 0.0.0.3 area 0
```

```
R3# conf t
(config)# router ospf 1
(config-router)# router-id 3.3.3.3
(config-router)# network 10.0.1.0 0.0.0.3 area 0
(config-router)# network 10.0.2.0 0.0.0.3 area 0
```

```
R1# conf t
(config)# router ospf 1
(config-router)# router-id 4.4.4.4
(config-router)# network 10.0.2.0 0.0.0.3 area 0
(config-router)# network 10.0.3.0 0.0.0.3 area 0
```

Inspection commands

Router# show ip route

Router# show ip ospf neighbor

- Configuration is bellow:

```
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#router ospf 1
R2(config-router)#router-id 1.1.1.1
R2(config-router)#network 10.0.0.0 0.0.0.3 area 0
R2(config-router)#network 10.0.3.0 0.0.0.3 area 0

Enter configuration commands, one per line. End with CNTL/Z.
R-MIDDLE(config)#router ospf 1
R-MIDDLE(config-router)#router-id 2.2.2.2
R-MIDDLE(config-router)#network 10.0.0.0 0.0.0.3 area 0
R-MIDDLE(config-router)#network 10.0.0.0 0.0.0.3 area 0
*Dec 17 11:44:45.547: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on GigabitEthernet0/0 f
rom LOADING to FULL, Loading Done
R-MIDDLE(config-router)#network 10.0.1.0 0.0.0.3 area 0
```

Figures. R2 and R-MIDDLE

```
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#router ospf 1
R3(config-router)#router-id 3.3.3.3
R3(config-router)#network 10.0.1.0 0.0.0.3 area 0
R3(config-router)#network 10.0.1.0 0.0.0.3 area 0
*Dec 17 11:47:16.755: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet1/0 f
rom LOADING to FULL, Loading Done
R3(config-router)#network 10.0.2.0 0.0.0.3 area 0
R3(config-router)#
```

```
R1#
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#router ospf 1
R1(config-router)#router-id 4.4.4.4
R1(config-router)#network 10.0.2.0 0.0.0.3 area 0
R1(config-router)#network 10.0.3.0 0.0.0.3 area 0
R1(config-router)#
```

Figures. R3 and R1

- Inspection commands are below using `show ip route` and `show ip ospf neighbor` :

```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, I - ISIS
a - application route
+ - replicated route, % - next hop override

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C 10.0.0.0/30 is directly connected, GigabitEthernet0/0
L 10.0.0.1/32 is directly connected, GigabitEthernet0/0
O 10.0.1.0/30 [110/2] via 10.0.0.2, 00:04:08, GigabitEthernet0/0
O 10.0.2.0/30 [110/2] via 10.0.3.1, 00:02:28, GigabitEthernet1/0
C 10.0.3.0/30 is directly connected, GigabitEthernet1/0
L 10.0.3.2/32 is directly connected, GigabitEthernet1/0

R2#show ip ospf neighbor
Neighbor ID Pri State Dead Time Address Interface
4.4.4.4 1 FULL/BDR 00:00:34 10.0.3.1 GigabitEthernet1/0
2.2.2.2 1 FULL/BDR 00:00:36 10.0.0.2 GigabitEthernet0/0
```

```
R-MIDDLE#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, I - ISIS
a - application route
+ - replicated route, % - next hop override

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C 10.0.0.0/30 is directly connected, GigabitEthernet0/0
C 10.0.0.2/32 is directly connected, GigabitEthernet0/0
C 10.0.1.0/30 is directly connected, GigabitEthernet1/0
L 10.0.1.1/32 is directly connected, GigabitEthernet1/0
O 10.0.2.0/30 [110/2] via 10.0.1.2, 00:03:26, GigabitEthernet1/0
O 10.0.3.0/30 [110/2] via 10.0.0.1, 00:05:49, GigabitEthernet0/0

R-MIDDLE#show ip ospf neighbor
Neighbor ID Pri State Dead Time Address Interface
3.3.3.3 1 FULL/BDR 00:00:30 10.0.1.2 GigabitEthernet1/0
1.1.1.1 1 FULL/DR 00:00:35 10.0.0.1 GigabitEthernet0/0
```

Figures. R2 and R-MIDDLE

```
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, I - ISIS
a - application route
+ - replicated route, % - next hop override

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
O 10.0.0.0/30 [110/2] via 10.0.1.1, 00:01:37, GigabitEthernet1/0
C 10.0.1.0/30 is directly connected, GigabitEthernet1/0
L 10.0.1.2/32 is directly connected, GigabitEthernet1/0
C 10.0.2.0/30 is directly connected, GigabitEthernet0/0
L 10.0.2.1/32 is directly connected, GigabitEthernet0/0
O 10.0.3.0/30 [110/2] via 10.0.2.2, 00:00:06, GigabitEthernet0/0

R3#show ip ospf neighbor
Neighbor ID Pri State Dead Time Address Interface
4.4.4.4 1 FULL/DR 00:00:35 10.0.2.2 GigabitEthernet0/0
2.2.2.2 1 FULL/DR 00:00:37 10.0.1.1 GigabitEthernet1/0
R3#
```

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, I - ISIS
a - application route
+ - replicated route, % - next hop override

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
O 10.0.0.0/30 [110/2] via 10.0.3.2, 00:01:30, GigabitEthernet0/0
O 10.0.1.0/30 [110/2] via 10.0.2.1, 00:01:00, GigabitEthernet1/0
C 10.0.2.0/30 is directly connected, GigabitEthernet1/0
C 10.0.2.2/32 is directly connected, GigabitEthernet1/0
C 10.0.3.0/30 is directly connected, GigabitEthernet0/0
L 10.0.3.1/32 is directly connected, GigabitEthernet0/0

R1#show ospf neighbor
Neighbor ID Pri State Dead Time Address Interface
1.1.1.1 1 FULL/DR 00:00:33 10.0.3.2 GigabitEthernet0/0
3.3.3.3 1 FULL/BDR 00:00:39 10.0.2.1 GigabitEthernet1/0
R1#
```

Figures. R3 and R1

- OSPF allows pinging all remote nodes:

```

1.1.1.1 1 FULL/DK 00.00.33 10.0.0.1 GigabitEthernet0
R-MIDDLE#ping 10.0.3.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.3.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 12/19/24 ms
R-MIDDLE#ping 10.0.2.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.2.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 16/27/32 ms
R-MIDDLE#

```

Figure. R-MIDDLE pinging R1 from 2 sides

2. Which interface you will select as the OSPF router ID and why?

I would select loopback `lo0` interface.

Router ID (RID) can be chosen by OSPF automatically on the highest IP address in the net and also it will prefer loopback interface. The last one is chosen based on assumption that this kind of interface will not go down — simply, stability. If a interface flaps, then my entire net with OSPF will not require to be restarted.

Why the highest IP? Because it allow predictability, so we can even manually set the highest ID, so that we would certainly now that the logic will select this one node.

Router ID is a part of LSAs (Link-state Advertisements — is a data packet with router information: RID, metrics, cost, bandwidth — it is shared between routers to enable OSPF learning)

3. What is the difference between advertising all the networks VS manual advertising (per interface or per subnet)? Which one is better?

Advertising all networks is simple and easy to make. You can set up advertisement for a very broad wildcard, so that all IPs that match it will be advertised. Only if you add new subnet or interface you update the config.

But this one has limitations of security: you can share those IPs, whose presence is not wanted..

```
# ALL
router ospf 1
network 0.0.0.0 255.255.255.255 area 0 // This enables OSPF on **ALL** interfaces
```

```
#MANUAL

interface GigabitEthernet0/0
ip ospf 1 area 0 // Enables OSPF on this specific interface only

router ospf 1
network 0.0.0.0 255.255.255.255 area 0
passive-interface default // Suppresses Hellos on ALL interfaces
no passive-interface GigabitEthernet0/0 // Re-enable only on interconnects
```

Manual can be done for the whole net but keep interface passive, so it will advertise net, but will be able to communicate only with passive-configured nodes.

Manual is done per an interface. Gives more control and stability as well, as security. Better for scalable systems, so you do not need to change ospf configuration.

Cisco recommends using Manual advertising.

4. If you have a static route in a router, how can you let your OSPF neighbors know about it?

Approve and show it on practice.

- Configuration for the case (R5 and R3)

```
# Configure static IP on R3 g2/0
```

```
R3# conf t
```

```
(config)# interface g2/0
```

```
(config-if)# ip address 10.0.4.1 255.255.255.252
```

```
(config-if)# no shut
```

```
(config-if)# exit
```

```
# Configure static route on R3 to R5 network
```

```
(config)# ip route 192.168.1.0 255.255.255.252 g2/0
```

```
# Configure static route on R5 to R3 network
```

```
R5# conf t
```

```
(config)# ip route 10.0.4.0 255.255.255.252 g0/0
```

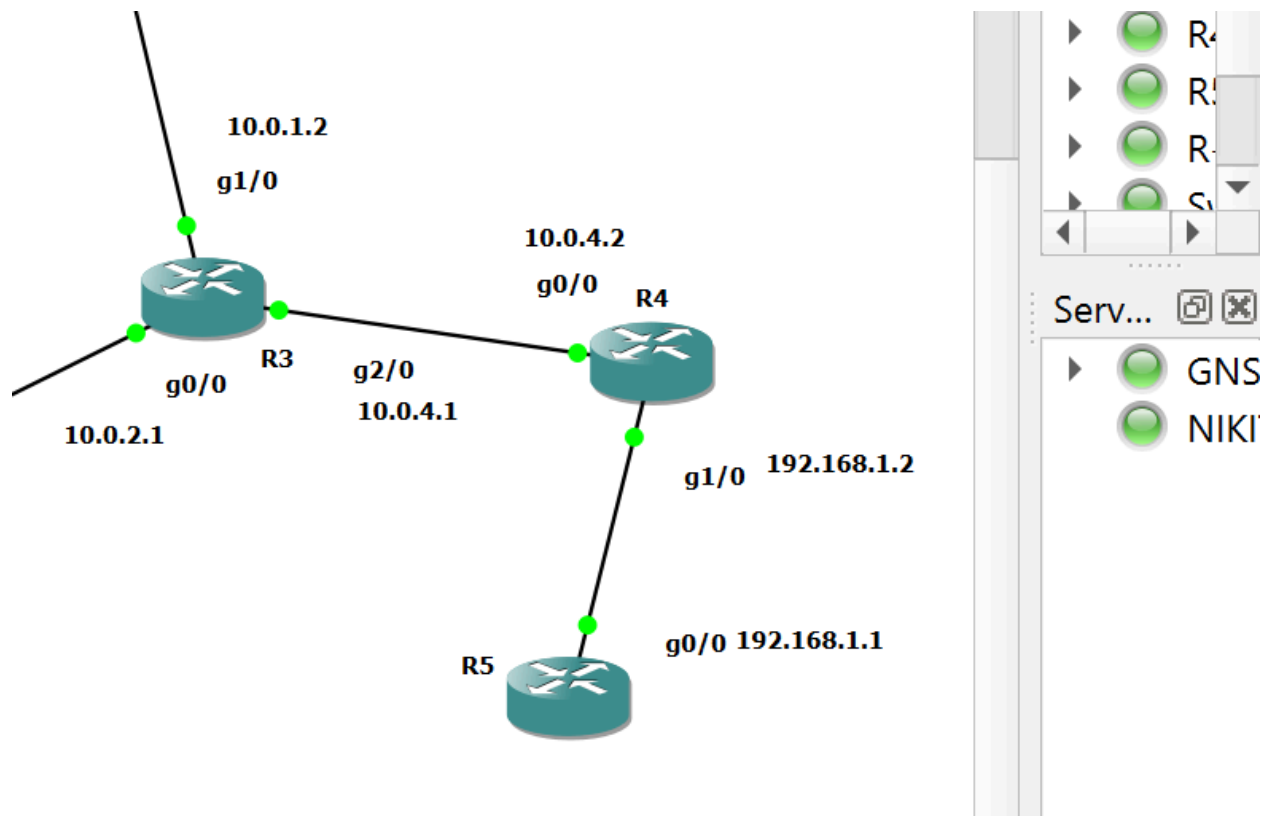


Figure. Case topology (frame)

- From R3 I branched 2 new routers, that will have a static route on network R4 $\leftrightarrow$ R5.
 - Configuration for the case (R4 and R5)

```
# Configure static IP on R3 g2/0
R3# conf t
(config)# interface g2/0
(config-if)# ip address 10.0.4.1 255.255.255.252
(config-if)# no shut
(config-if)# exit

# Configure static route on R3 to R5 network
(config)# ip route 192.168.1.0 255.255.255.252 g2/0

# Configure static route on R5 to R3 network
```

```
R5# conf t
(config)# ip route 10.0.4.0 255.255.255.252 g0/0
```

```
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#interface g2/0
R3(config-if)#ip address 10.0.4.1 255.255.255.252
R3(config-if)#no shut
R3(config-if)#exit
*Dec 17 12:21:43.535: %LINK-3-UPDOWN: Interface GigabitEthernet0/2/0, changed state to up
R3(config-if)#exit
R3(config)#ip route 192.168.1.0 255.255.255.252 g2/0

R5#
R5#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R5(config)#ip route 10.0.4.0 255.255.255.252 g0/0
```

Figure. R3 and R5

- To test accessibility I used ping (R3 $\leftrightarrow$ R5; R3 $\rightarrow$ R4)

```
R3#ping 192.168.1.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.2, timeout is 2 seconds:
.....
Success rate is 80 percent (4/5), round-trip min/avg/max = 8/14/20 ms
R3#ping 192.168.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.1, timeout is 2 seconds:
.....
Success rate is 100 percent (5/5), round-trip min/avg/max = 12/24/28 ms
R3#

R5#
R5#ping 10.0.4.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.4.1, timeout is 2 seconds:
.....
Success rate is 100 percent (5/5), round-trip min/avg/max = 20/27/32 ms
```

Figure. R3, R5 ping on static routes

```

R3# show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
O       10.0.0.0/30 [110/2] via 10.0.1.1, 00:40:36, GigabitEthernet1/0
C       10.0.1.0/30 is directly connected, GigabitEthernet1/0
L       10.0.1.2/32 is directly connected, GigabitEthernet1/0
C       10.0.2.0/30 is directly connected, GigabitEthernet0/0
L       10.0.2.1/32 is directly connected, GigabitEthernet0/0
O       10.0.3.0/30 [110/2] via 10.0.2.2, 00:39:05, GigabitEthernet0/0
C       10.0.4.0/30 is directly connected, GigabitEthernet2/0
L       10.0.4.1/32 is directly connected, GigabitEthernet2/0
    192.168.1.0/30 is subnetted, 1 subnets

```

Figure. Subnets on R3

<pre> R5(config)#interface g0/0 R5(config-if)#ip address 192.168.1.1 255.255.255.252 R5(config-if)#no shut </pre>	<pre> R4#conf t Enter configuration commands, one per line. End with R4(config)#interface g1/0 R4(config-if)#ip address 192.168.1.2 255.255.255.252 R4(config-if)#no shut R4(config-if)#i *Dec 17 12:21:19.531: %LINK-3-UPDOWN: Interface Giga up *Dec 17 12:21:20.531: %LINEPROTO-5-UPDOWN: Line prot t1/0, changed state to up R4(config-if)#interface g/0 ^ % Invalid input detected at '^' marker. R4(config)#interface g0/0 R4(config-if)#ip address 10.0.4.2 % Incomplete command. R4(config-if)#ip address 10.0.4.2 255.255.255.252 R4(config-if)#no shut </pre>
---	---

Figure. R5 and R4 configuration of static IP addresses

- Right configuration will allow to **let others find** those “guys” and ping the new static net without any manual configuration - learning process takes it place.
!!! R6 will not correctly reply to unconfigured routes for those other pinging nodes.

```
# Let R3 people know about route to guys on R3-R4 network (10.0.4.0/30)
)
R3(config)# router ospf 1
R3(config-router)# network 10.0.4.0 0.0.0.3 area 0
# Redistribute static subnets
(config-router)# redistribute static subnets
```

```
Enter configuration commands, one per line. End with
R3(config)#router ospf 1
R3(config-router)# network 10.0.4.0 0.0.0.3 area 0
R3(config-router)#redistribute static subnets
```

Figure. R3 configuration

- Verification

<pre>R1#show ip route Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2 E1 - OSPF external type 1, E2 - OSPF external type 2 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2 ia - IS-IS inter area, * - candidate default, U - per-user static route o - ODR, P - periodic downloaded static route, H - NHRP, 1 - LISP a - application route + - replicated route, % - next hop override Gateway of last resort is not set 10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks O 10.0.0.0/30 [110/2] via 10.0.3.2, 00:47:42, GigabitEthernet0/0 O 10.0.1.0/30 [110/2] via 10.0.2.1, 00:47:12, GigabitEthernet1/0 C 10.0.2.0/30 is directly connected, GigabitEthernet1/0 L 10.0.2.2/32 is directly connected, GigabitEthernet1/0 C 10.0.3.0/30 is directly connected, GigabitEthernet0/0 L 10.0.3.1/32 is directly connected, GigabitEthernet0/0 O 10.0.4.0/30 [110/2] via 10.0.2.1, 00:01:00, GigabitEthernet1/0 O E2 192.168.1.0/30 is subnetted, 1 subnets O E2 192.168.1.0 [110/20] via 10.0.2.1, 00:00:29, GigabitEthernet1/0 R1#</pre>	<pre>R2#show ip route Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2 E1 - OSPF external type 1, E2 - OSPF external type 2 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2 ia - IS-IS inter area, * - candidate default, U - per-user static route o - ODR, P - periodic downloaded static route, H - NHRP, 1 - LISP a - application route + - replicated route, % - next hop override Gateway of last resort is not set 10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks C 10.0.0.0/30 is directly connected, GigabitEthernet0/0 L 10.0.0.1/32 is directly connected, GigabitEthernet0/0 O 10.0.1.0/30 [110/2] via 10.0.0.2, 00:51:43, GigabitEthernet0/0 O 10.0.2.0/30 [110/2] via 10.0.3.1, 00:50:03, GigabitEthernet1/0 C 10.0.3.0/30 is directly connected, GigabitEthernet1/0 L 10.0.3.2/32 is directly connected, GigabitEthernet1/0 O 10.0.4.0/30 [110/3] via 10.0.3.1, 00:03:19, GigabitEthernet1/0 [110/3] via 10.0.0.2, 00:03:19, GigabitEthernet0/0 O E2 192.168.1.0/30 is subnetted, 1 subnets O E2 192.168.1.0 [110/20] via 10.0.3.1, 00:03:19, GigabitEthernet1/0 R2#</pre>
---	--

```

Success rate is 0 percent (0/5)
R-MIDDLE#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0
L       10.0.0.2/32 is directly connected, GigabitEthernet0/0
C       10.0.1.0/30 is directly connected, GigabitEthernet1/0
L       10.0.1.1/32 is directly connected, GigabitEthernet1/0
O       10.0.2.0/30 [110/2] via 10.0.1.2, 00:53:31, GigabitEthernet1/0
O       10.0.3.0/30 [110/2] via 10.0.0.1, 00:55:54, GigabitEthernet0/0
O       10.0.4.0/30 [110/2] via 10.0.1.2, 00:07:19, GigabitEthernet1/0
    192.168.1.0/30 is subnetted, 1 subnets
O E2    192.168.1.0 [110/20] via 10.0.1.2, 00:06:48, GigabitEthernet1/0
R-MIDDLE#

```

Figure. Static net is seen on all other routers (e.g. R1,R2) outside this statnet.

5. Enable OSPF with authentication between the neighbours and verify it.
 - I will enable recommended measures as setting up MD5 hash auth (password) on desired neighbours (R-MIDDLE)

```

R-MIDDLE#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R-MIDDLE(config)#interface range g0/0, g1/0
R-MIDDLE(config-if-range)#ip ospf authen
R-MIDDLE(config-if-range)#ip ospf authentication message-digest
R-MIDDLE(config-if-range)#ip ospf message-digest-key 1 md5 ospf-pass
R-MIDDLE(config-if-range)#ip ospf 1 area 0
R-MIDDLE(config-if-range)#

```

Figure. R-MIDDLE config for security

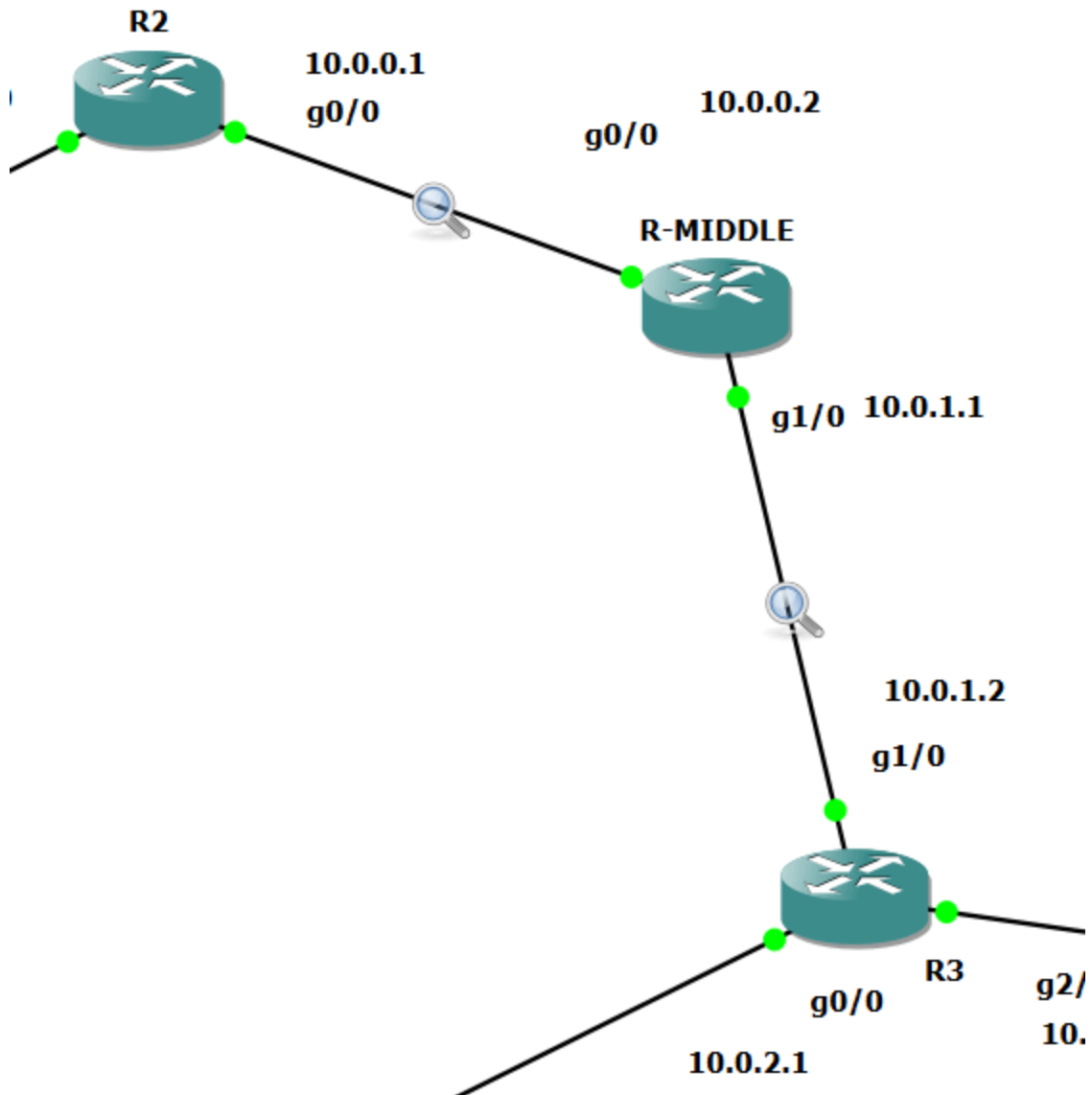


Figure. Starting capture mode

- Now authentication mechanism are seen in packets send (headers)
 - Auth Type: Cryptographic (2)
 - Auth Crypt Data: 3867cbe8cfc7e9e8e49b904f98048b38
 - ...

Захват из Standard input [R-MIDDLE GigabitEthernet0/0 to R2 GigabitEthernet0/0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

Примените фильтр отображения ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
3	0.548884	10.0.0.2	224.0.0.5	OSPF	130	Hello Packet
4	2.361227	ca:02:07:ed:00:08	CDP/VTP/DTP/PagP/UD...	CDP	371	Device ID: R2 Port ID: GigabitEthernet0/0
5	2.492579	10.0.0.1	224.0.0.5	OSPF	90	Hello Packet
6	7.383630	ca:04:08:29:00:08	CDP/VTP/DTP/PagP/UD...	CDP	377	Device ID: R-MIDDLE Port ID: GigabitEthernet0/0
7	9.954674	10.0.0.2	224.0.0.5	OSPF	130	Hello Packet
8	9.992006	ca:02:07:ed:00:08	ca:02:07:ed:00:08	LOOP	60	Reply
9	10.116807	ca:04:08:29:00:08	ca:04:08:29:00:08	LOOP	60	Reply
10	12.174871	10.0.0.1	224.0.0.5	OSPF	90	Hello Packet
11	19.331359	10.0.0.2	224.0.0.5	OSPF	130	Hello Packet
12	20.008114	ca:02:07:ed:00:08	ca:02:07:ed:00:08	LOOP	60	Reply
13	20.138862	ca:04:08:29:00:08	ca:04:08:29:00:08	LOOP	60	Reply
14	21.645428	10.0.0.1	224.0.0.5	OSPF	90	Hello Packet
15	29.331337	10.0.0.2	224.0.0.5	OSPF	130	Hello Packet
16	30.014086	ca:02:07:ed:00:08	ca:02:07:ed:00:08	LOOP	60	Reply
17	30.139578	ca:04:08:29:00:08	ca:04:08:29:00:08	LOOP	60	Reply
18	30.791803	10.0.0.1	224.0.0.5	OSPF	90	Hello Packet
19	38.865609	10.0.0.2	224.0.0.5	OSPF	130	Hello Packet
20	40.006166	ca:02:07:ed:00:08	ca:02:07:ed:00:08	LOOP	60	Reply
21	40.107065	10.0.0.1	224.0.0.5	OSPF	90	Hello Packet
22	40.119346	ca:04:08:29:00:08	ca:04:08:29:00:08	LOOP	60	Reply
23	47.891106	10.0.0.2	224.0.0.5	OSPF	130	Hello Packet
24	49.678186	10.0.0.1	224.0.0.5	OSPF	90	Hello Packet
25	49.991204	ca:02:07:ed:00:08	ca:02:07:ed:00:08	LOOP	60	Reply
26	50.114220	ca:04:08:29:00:08	ca:04:08:29:00:08	LOOP	60	Reply

..0. = More fragments: Not set
 ...0 0000 0000 0000 = Fragment Offset: 0
 Time to Live: 1
 Protocol: OSPF IGP (89)
 Header Checksum: 0xcb07 [validation disabled]
 [Header checksum status: Unverified]
 Source Address: 10.0.0.2
 Destination Address: 224.0.0.5
 [Stream index: 0]

Open Shortest Path First

- OSPF Header
 - Version: 2
 - Message Type: Hello Packet (1)
 - Packet Length: 44
 - Source OSPF Router: 2.2.2.2
 - Area ID: 0.0.0.0 (Backbone)
 - Checksum: 0x0000 (None)
 - Auth Type: Cryptographic (2)
 - Auth Crypt Key id: 1
 - Auth Crypt Data Length: 16
 - Auth Crypt Sequence Number: 1765975514
 - Auth Crypt Data: 3867cbe8cfc7e9e8e49b904f98048b38
- OSPF Hello Packet
- OSPF LLS Data Block

0000 01 00 5e 00 00 05 ca 04 08 29 00 08 08 00 45 c0
 0010 00 74 03 63 00 00 01 59 cb 07 0a 00 00 02 e0 00
 0020 00 05 02 01 00 2c 02 02 02 02 00 00 00 00 00
 0030 00 02 00 00 01 10 69 42 a5 da ff ff ff fc 00 0a
 0040 12 01 00 00 00 28 0a 00 00 02 00 00 00 00 38 67
 0050 cb e8 cf c7 e9 e8 e4 9b 90 4f 98 04 8b 38 00 00
 0060 00 09 00 01 00 04 00 00 00 01 00 02 00 14 69 42
 0070 a5 da b4 64 6a 8f 0f 37 ab 35 ce b4 bd 1a d0 b9
 0080 9a d6

Figure. R-MIDDLE $\leftrightarrow$ R3: Hello Packets are authenticated (Header)

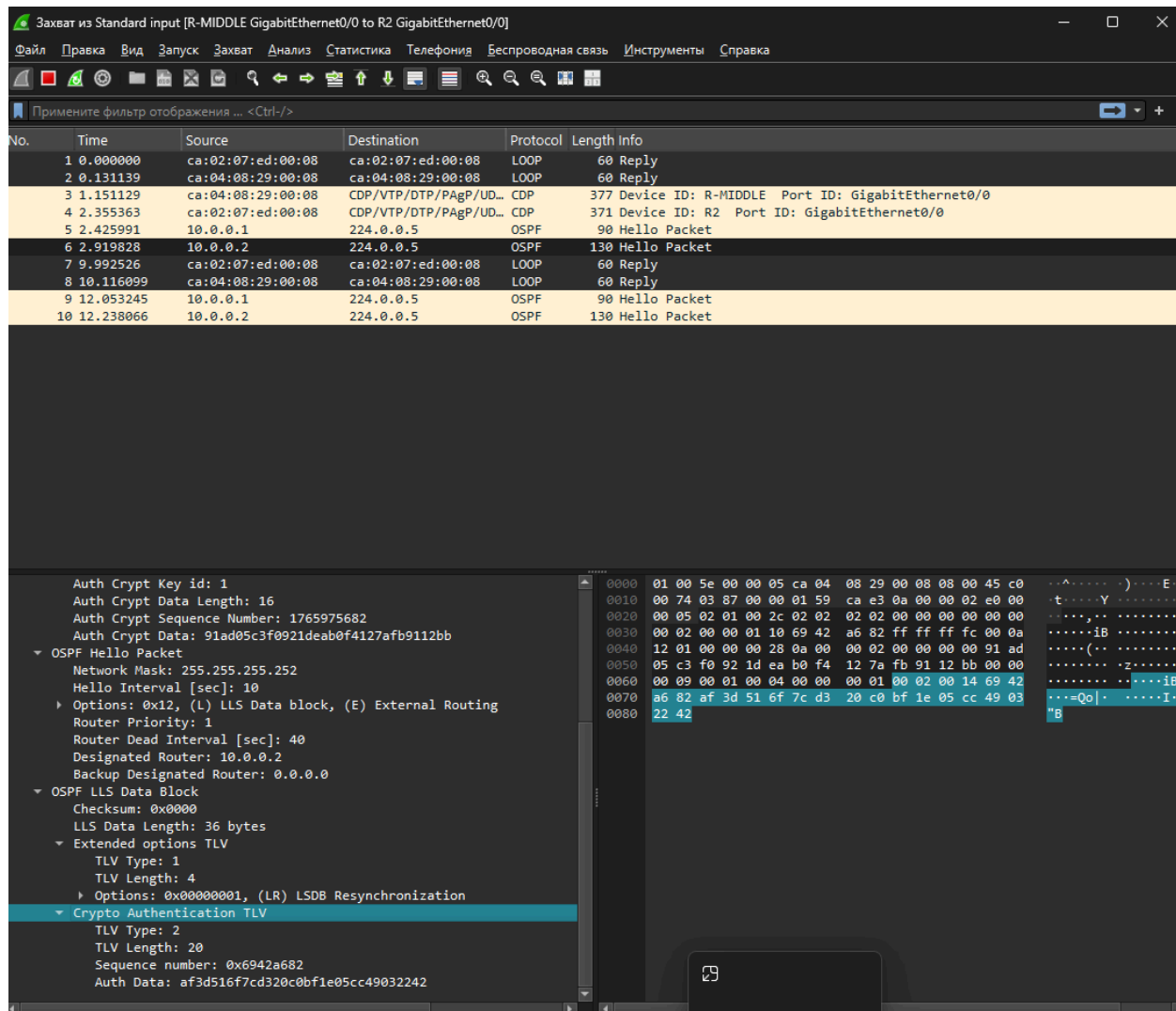


Figure. R-MIDDLE ↔ R2: Hello Packets are authenticated (Data block)

Task 3 - OSPF Verification

1. How can you check if you have a full adjacency with your router neighbour?

`show ip ospf neighbor`

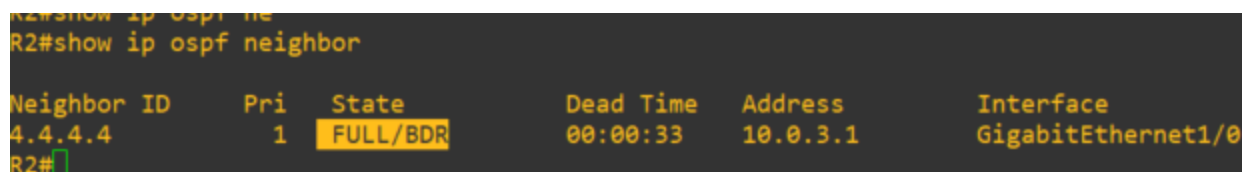


Figure. full adjacency state on R2 with his neighbour R-MIDDLE

If adjacency is incomplete, then state = EXSTART, EXCHANGE, LOADING, etc.,
(common issues: MTU mismatch, authentication failure, network type mismatch).

`debug ip ospf adj` to see adjacency progression.

```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

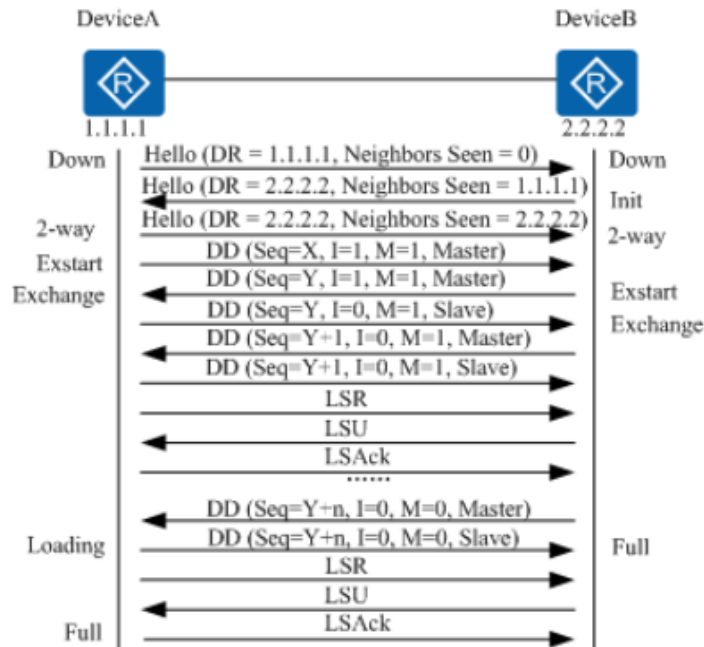
Gateway of last resort is 192.168.0.5 to network 0.0.0.0

R   10.0.0.0/8 [120/1] via 192.168.0.10 00:00:05, GigabitEthernet0/2
C   20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C   20.0.0.0/8 is directly connected, GigabitEthernet0/0
L   20.0.0.1/32 is directly connected, GigabitEthernet0/0
R   30.0.0.0/8 [120/1] via 192.168.0.5 00:00:05, GigabitEthernet0/1
R   192.168.0.0/24 is variably subnetted, 5 subnets, 2 masks
R   192.168.0.0/30 [120/1] via 192.168.0.5 00:00:05, GigabitEthernet0/1
R   192.168.0.0/30 [120/1] via 192.168.0.10 00:00:05, GigabitEthernet0/2
C   192.168.0.4/30 is directly connected, GigabitEthernet0/1
L   192.168.0.6/32 is directly connected, GigabitEthernet0/1
C   192.168.0.8/30 is directly connected, GigabitEthernet0/2
L   192.168.0.9/32 is directly connected, GigabitEthernet0/2
S*  0.0.0.0/0 [1/0] via 192.168.0.5
```

Diagram annotations on the route table:

- AD** (Administrative Distance) points to the first number in the route table (e.g., 120).
- Metric** points to the second number in the route table (e.g., 1).
- Elapsed time** points to the time value (e.g., 00:00:05).
- Exit interface** points to the interface name (e.g., GigabitEthernet0/2).
- Next hop IP address** points to the IP address in the 'via' field (e.g., 192.168.0.5).

Figure. Scheme



Adjacency establishment process on a broadcast network

Figure. Adjacency establishment process on a broadcast network

"The adjacency establishment process is as follows:

1. The local and remote routers use OSPF interfaces to exchange Hello packets to establish a neighbour relationship.
2. The local and remote routers negotiate a master/slave relationship and exchange Database Description (DD) packets.
3. The local and remote routers exchange link state advertisements (LSAs) to synchronize their link state databases (LSDBs)"

2. How can you check in the routing table which networks did you receive from your neighbours?

`show ip route ospf` — shows only received networks by OSPF from guys in the net.

First code from the table indicates from which type of neighbours table was learned, e.g. R — by RIP protocol, B — by BGP protocol..

```
R2#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
O       10.0.1.0/30 [110/3] via 10.0.3.1, 00:28:17, GigabitEthernet1/0
O       10.0.2.0/30 [110/2] via 10.0.3.1, 01:27:53, GigabitEthernet1/0
O       10.0.4.0/30 [110/3] via 10.0.3.1, 00:41:09, GigabitEthernet1/0
    192.168.1.0/30 is subnetted, 1 subnets
O E2    192.168.1.0 [110/20] via 10.0.3.1, 00:40:38, GigabitEthernet1/0
R2#
```

Figure. Route table on R2

3. Use traceroute to verify that you have a full OSPF network.

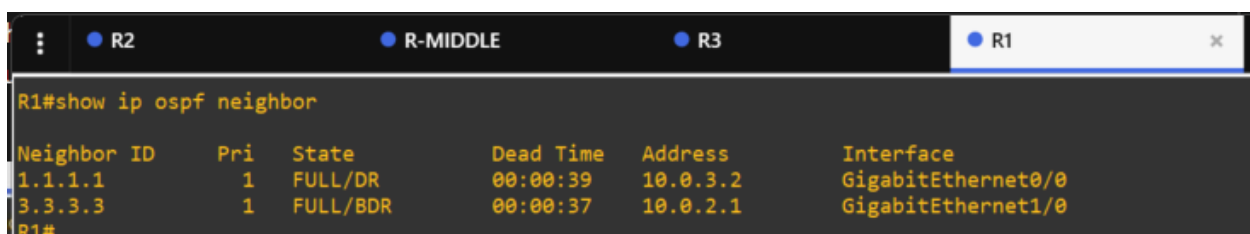
```
Tracing the route to 10.0.4.1
VRF info: (vrf in name/id, vrf out name/id)
  0  10.0.3.1 12 msec 16 msec 12 msec
  1  10.0.2.1 20 msec 40 msec 28 msec
R2#traceroute 10.0.1.2
Type escape sequence to abort.
Tracing the route to 10.0.1.2
VRF info: (vrf in name/id, vrf out name/id)
  0  10.0.3.1 12 msec 16 msec 20 msec
  1  10.0.2.1 16 msec 36 msec 40 msec
R2#
```

4. Which router is selected as DR and which one is BDR ?

They are selected by all routers in the topology. DR priority is used to pick between elected DR candidates. DR and BDR are chosen at the same time. Routers share LSAs with DR and BDR only so less bandwidth and adjacency. If DR

fails, BDR takes his place and becomes DR and all routers start to elect new BDR, whilst not affecting routing speed.

FULL/DR and FULL/BDR — 2 type of states indicating current role of a shown router. DR — designated router, BDR — backup designated routes. FULL — stands for complete sync and OSPF set up between routers states. R-MIDDLE is fully adjacent and is designated for accessing the network.



Neighbor ID	Pri	State	Dead Time	Address	Interface
1.1.1.1	1	FULL/DR	00:00:39	10.0.3.2	GigabitEthernet0/0
3.3.3.3	1	FULL/BDR	00:00:37	10.0.2.1	GigabitEthernet1/0

Figure. R1 table of neighbours

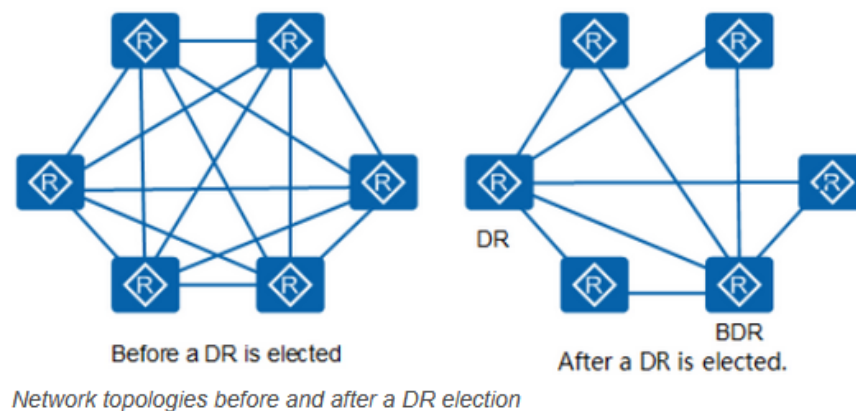


Figure. [Link](#)

5. Check what is the cost for each network that has been received by OSPF in the routing table.

Figure in question 2 shows cost and metric:

O [110/3], O [110/2], O [110/3], E [110/20]

Intra-area is O, External is E.

priority is 110 by default for OSPF and metric is the path cost.

- Cost are 2/3 for regular (O) routes. Difference between is the quantity of hops
- Cost is 20 for external (E2) routes.